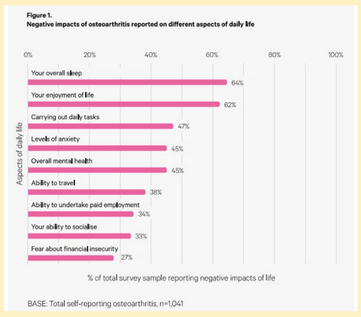
CONCEPTUALISATION OF A CHILD-PROOF OPENING DEVICE FOR THE MOVEMENT IMPAIRED
ME20018 Design 3: Product Design

DESIGN STATEMENT: Design a simple / easy to use product to decrease the input force required for opening child-proof bottles

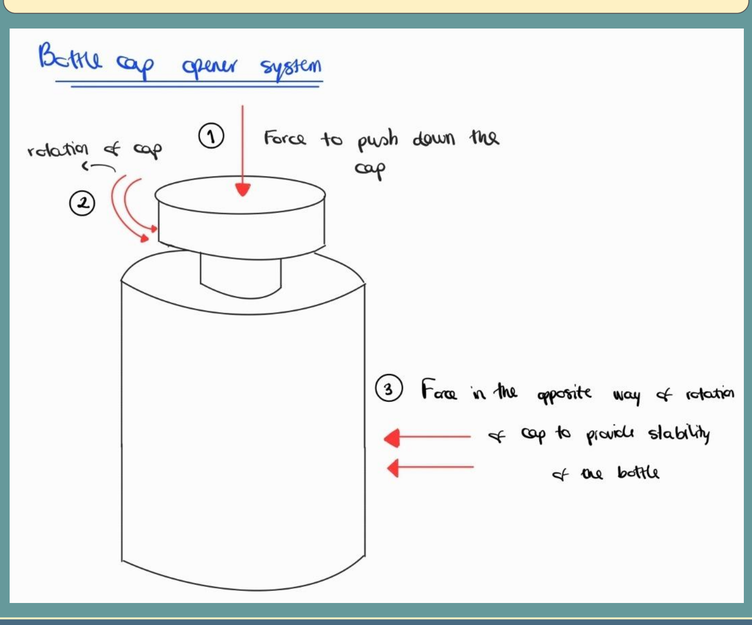
OSTEOARTHRITIS

Osteoarthritis is a degenerative joint disease, it occurs due to cartilage wearing - 7% of the total population develops osteoarthritis (around 500 million people). It is usually developed in woman after menopause, around the age of 50.

Symptoms:
- Limited movement in joints.
- Pain when using the join.
- Feeling that the joint is loose.



FORCE ANALYSIS



WORKING PRINCIPLE

1. User holds lid down with force F.
2. User grips onto lid, lid exerts a reaction force N.
3. User rotates hand with linear force x, centre of rotation either wrist or elbow with distance r to lid.
4. Force μN prevents hand from slipping, allows transmission of torque.
5. Torque applied onto lid T = xr.
6. Counter-rotation R applied to bottle to oppose T.
7. T > R, therefore lid comes loose.

For easier opening, x must be decreased, T must be increased by increasing μ, N and r.

EXISTING PRODUCT INSPIRATION

1. MULTI GRIP CAP OPENER

The Safety Cap Opener features wings that give it a stronger grip, allowing to turn the cap with ease



Disadvantages: Not suitable for bottles of different sizes. Despite the wings giving extra grip, still does not open

janice bell
★☆☆☆☆ **Very much a disappointing purchase**
Reviewed in the United Kingdom on 21 April 2021
Verified Purchase
This product isn't very good as it struggles to open safety caps and even normal bottle screw caps and leaves jagged bits on the safety lids...ive actually thrown the opener in the bin as it was no use whatsoever

jean timmins
★☆☆☆☆ **Cap opener, not suitable for purpose**
Reviewed in the United Kingdom on 30 September 2022
Verified Purchase
This is not suitable for smaller bottles. It is not suitable for purpose. It cannot be returned!! Not suitable for weak, arthritic hands.

2. MULTI PURPOSE LID OPENER



- The adjustable clamp fits around most awkward tops and jars, and requires only minimal squeezing pressure to operate.
- Limited to only twist caps and requires additional strength to clamp the top to the lid, making it difficult for people with osteoarthritis.
- Does not have a component at the base to hold the jar, thus making it difficult for people with osteoarthritis to grip, especially for bottles with larger diameters.

3. MASTER OPENER



- All in one opener mostly used for cans, ring-pulls, etc.
- Biggest disadvantage is that, even if it makes the lid loose, there is still considerable force required to completely open the jar/bottle.
- Also wears out quickly after some use. Therefore, not economical.

Requirements Table

DESIGN REQUIREMENTS / PRODUCT SPECIFICATION

Group Name	Requirements List *For a child-proof bottle cap opener*	Issued on: 20 / 11 / 2023	
Demands / Wants	*Requirement*	*Name*	*Changes (date)*
	1) Performance		
D	The bottle opener shall provide mechanical advantage when applying a torque to open the cap	lp2022	20/11/2023
D	The bottle opener shall not require any additional help from an abled person to operate	lp2022	20/11/2023
W	The bottle opener shall be simple and intuitive to use requiring little force to use	lp2022, ljc209	21/11/2023
D	The system shall work for bottles with different sizes	nf465	21/11/2023
W	The bottle opener shall be available at an accessible price point	lp2022	20/11/2023
	2) Geometry		
W	The bottle opener shall have a mass no more than 700 grams	lp2022	21/11/2023
W	The system shall minimize the number of components	nf4654	21/11/2023
W	The bottle opener shall be easy to store	ljc209	21/11/2023
W	There would be a base holder to grip the bottle for ease of the user	nf465	28/11/2023
	3) Safety		
W	The mechanism should not have sharp edges	nf465	21/11/2023
W	If the product falls/is dropped, it does not cause serious injuries due to its weight	nf465	21/11/2023
D	The holding system should not be stiff when used, so as to cause pain/injury.	nf465	21/11/2023
	4) Material		
W	The handle's material should be comfortable to grip onto	lp2022	21 / 11 / 2023
W	The materials used should be sustainable/eco-friendly	nf465	21/11/2023
	5) Ergonomics		
W	The device's handle shall be easy to grab	lp2022	21 / 11 / 2023

Problem Statement

Design an easy to grip device to make the opening of child-proof bottle caps accessible and easier for people with osteoarthritis.

References

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